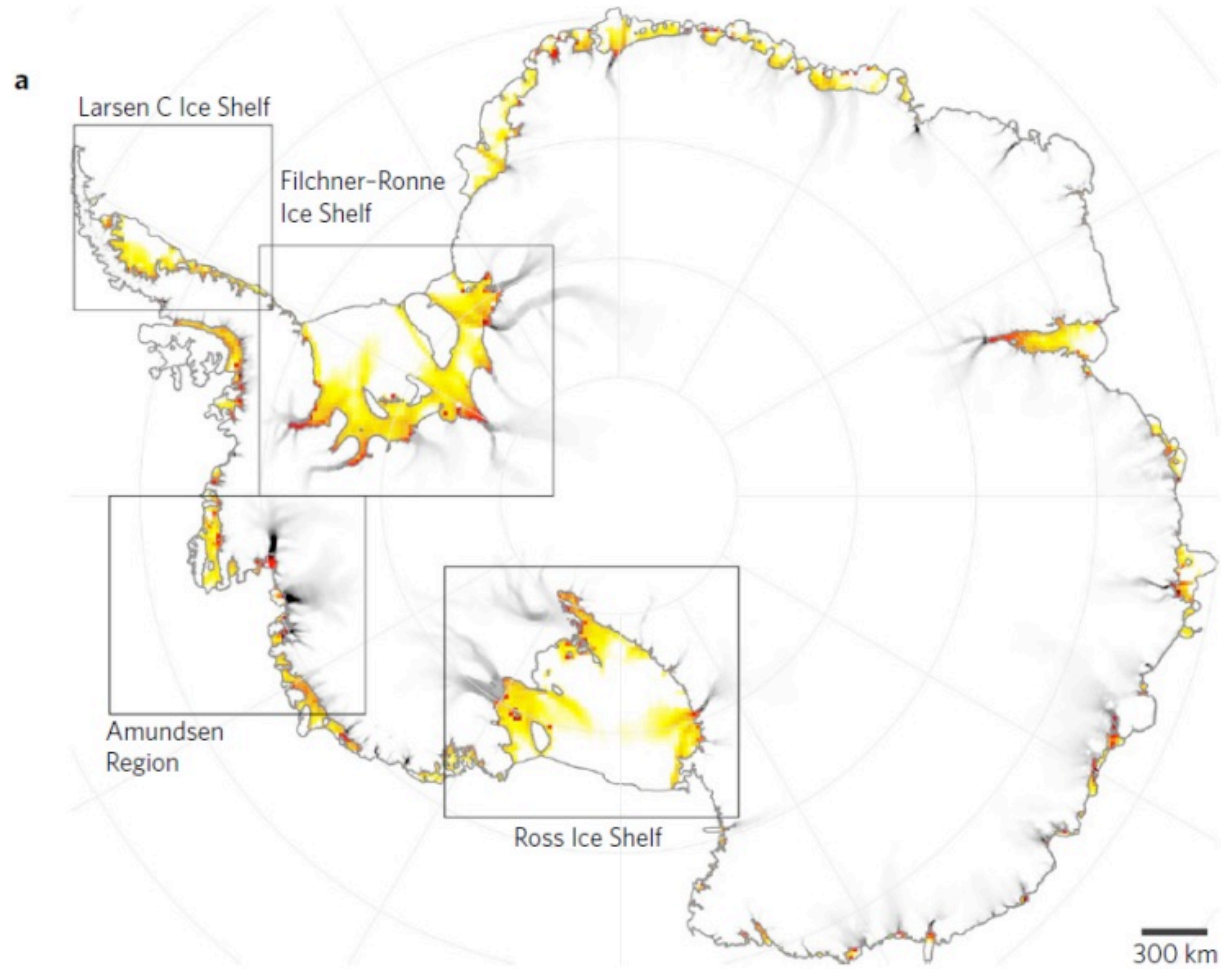


Lecture 9: Review + Ice Shelves

Review of first ½ of semester

Ice shelves

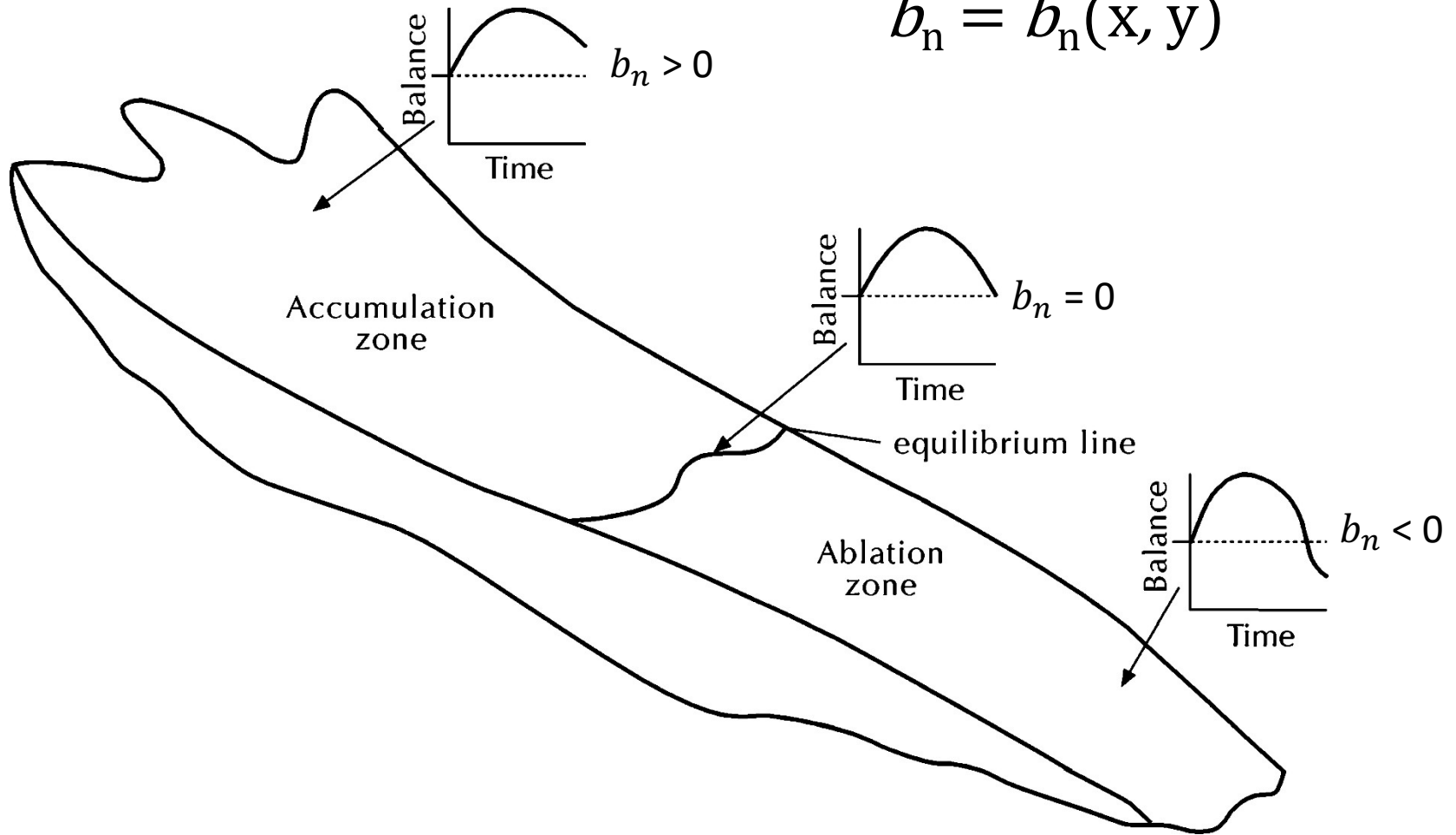


Review of first half of semester

- Surface mass balance
 - Accumulation zone processes
 - Surface energy balance
 - Basal melting
- Ice flow
 - Rheology
 - Stress balance
 - SIA model
 - Basal motion
 - Ice divides
 - Grounding lines

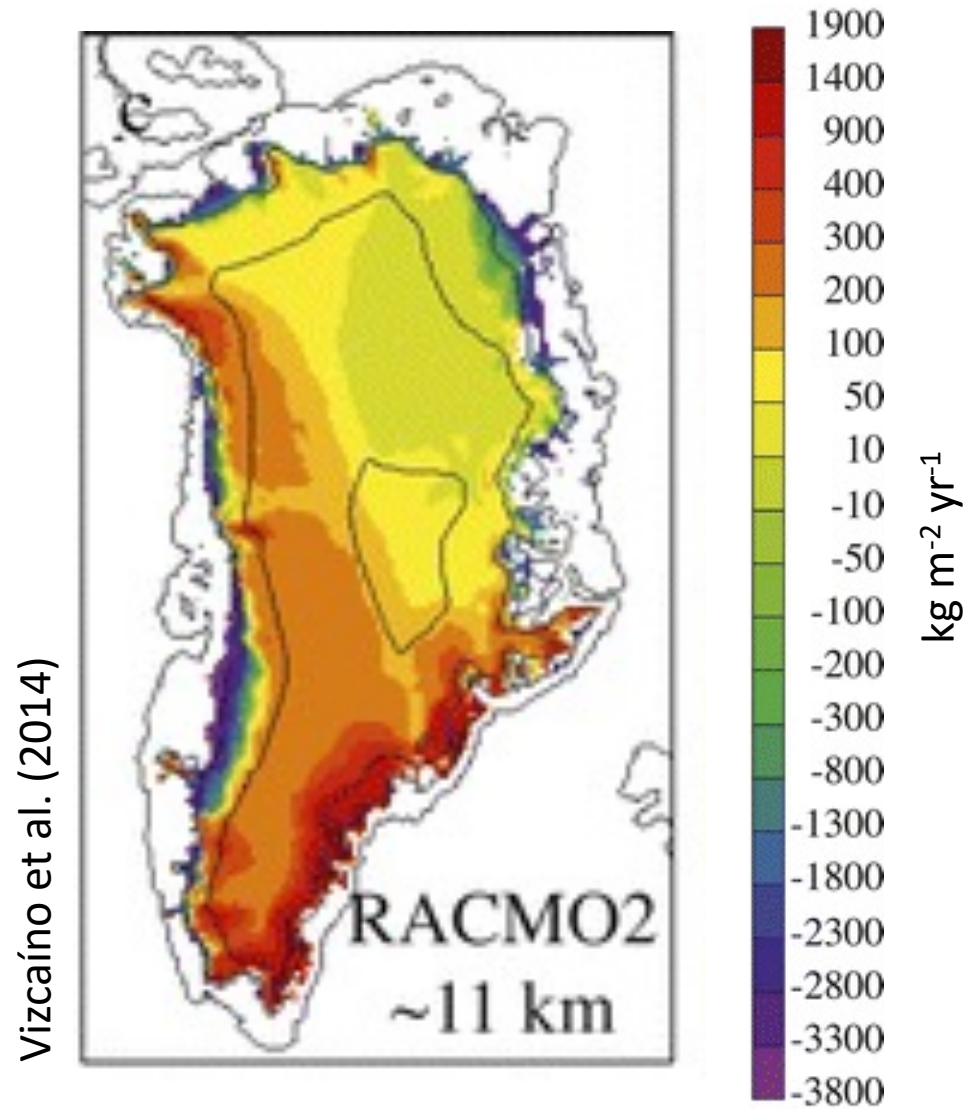
Spatially-varying net specific balance

$$b_n = b_n(x, y)$$



Cuffey and Paterson (2010)

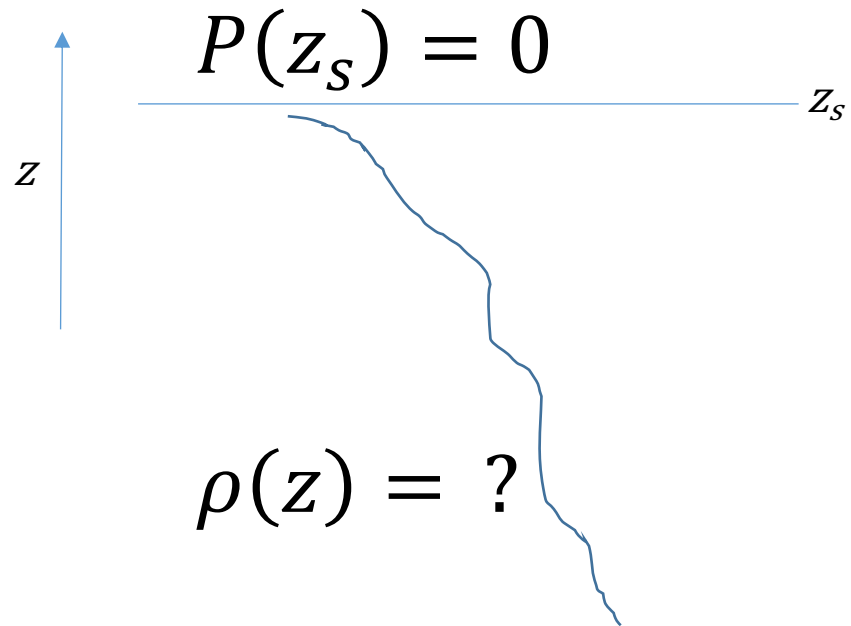
Long-term SMB of Greenland



Overburden pressure

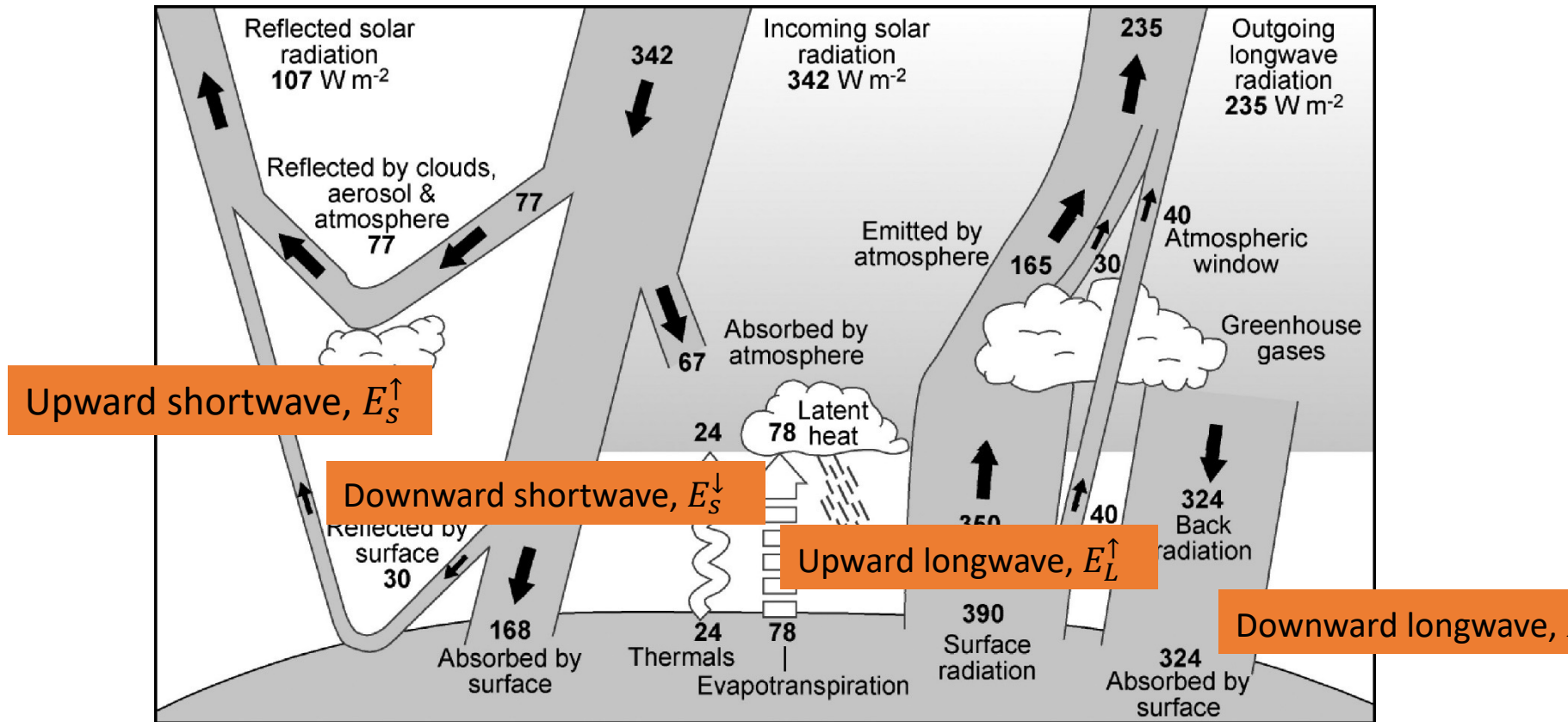
We want an equation for how pressure P varies with depth.

$$\frac{dP}{dz} = \rho(z)g$$



Surface energy balance

$$E_N = E_S^{\downarrow} + E_S^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_H + E_E$$



Stress, strain, rheology, mass balance

$$\dot{\epsilon} = \begin{bmatrix} \dot{\epsilon}_{xx} & \dot{\epsilon}_{xy} & \dot{\epsilon}_{xz} \\ \dot{\epsilon}_{yx} & \dot{\epsilon}_{yy} & \dot{\epsilon}_{yz} \\ \dot{\epsilon}_{zx} & \dot{\epsilon}_{zy} & \dot{\epsilon}_{zz} \end{bmatrix}$$

$$\sigma = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$

$$\epsilon_{ij} = \frac{1}{2} \left[\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right]$$

$$\sigma = \tau + \sigma_m$$

$$\dot{\epsilon}_{ij} = A \tau_E^{n-1} \tau_{ij} \quad (i,j = x, y, z)$$

$$\tau_E^2 = \frac{1}{2} [\tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2] + \tau_{xz}^2 + \tau_{xy}^2 + \tau_{yz}^2$$

$$\frac{\partial H}{\partial t} + \frac{\partial q}{\partial x} = b_i$$

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} = 0$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = \rho g$$

Ice flux in the Shallow Ice Approximation

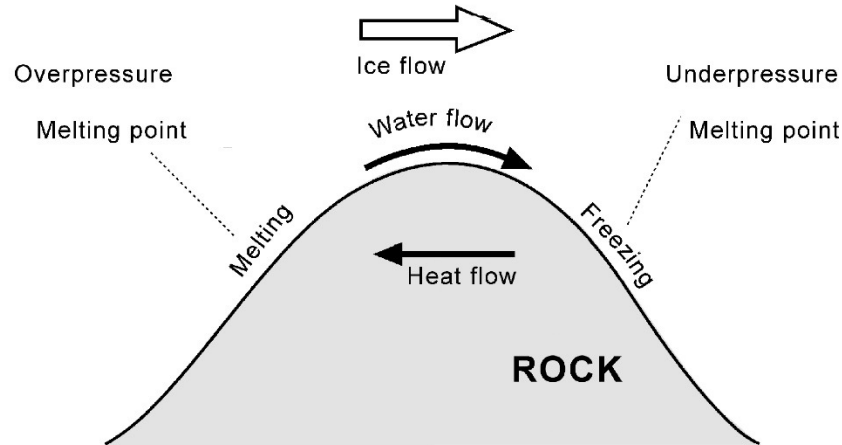
Depth-integrated horizontal velocity per unit width:

$$\hat{u} = 2A(\rho g \alpha H)^n \left(\frac{H}{n+2} \right)$$

Depth-integrated volume-flux per unit width:

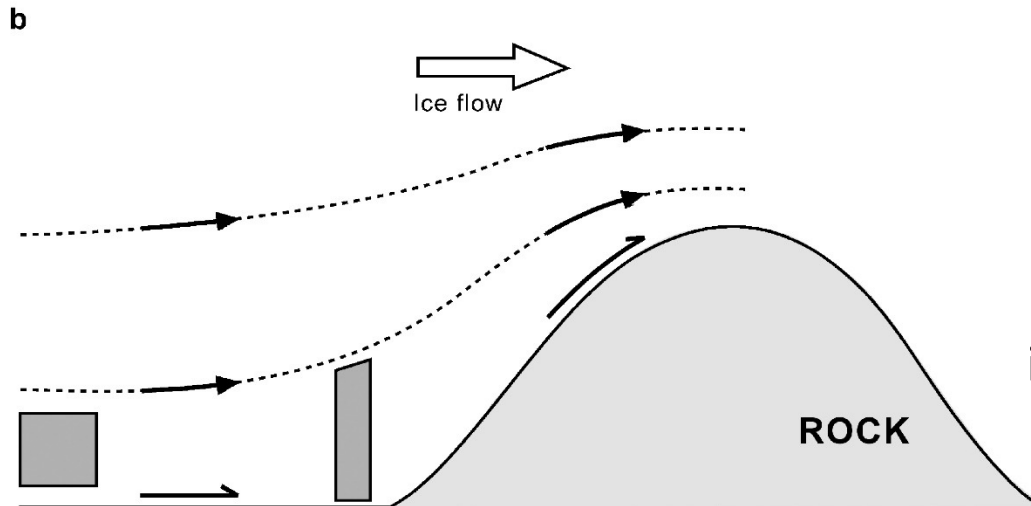
$$q = \hat{u}H = \frac{2A}{n+2} (\rho g \alpha)^n H^{n+2}$$

Weertman's theory for hard beds



Two mechanisms:
Regelation & enhanced creep.

$$u_b \propto \left(\frac{\sqrt{\tau_b}}{R} \right)^{n+1}$$

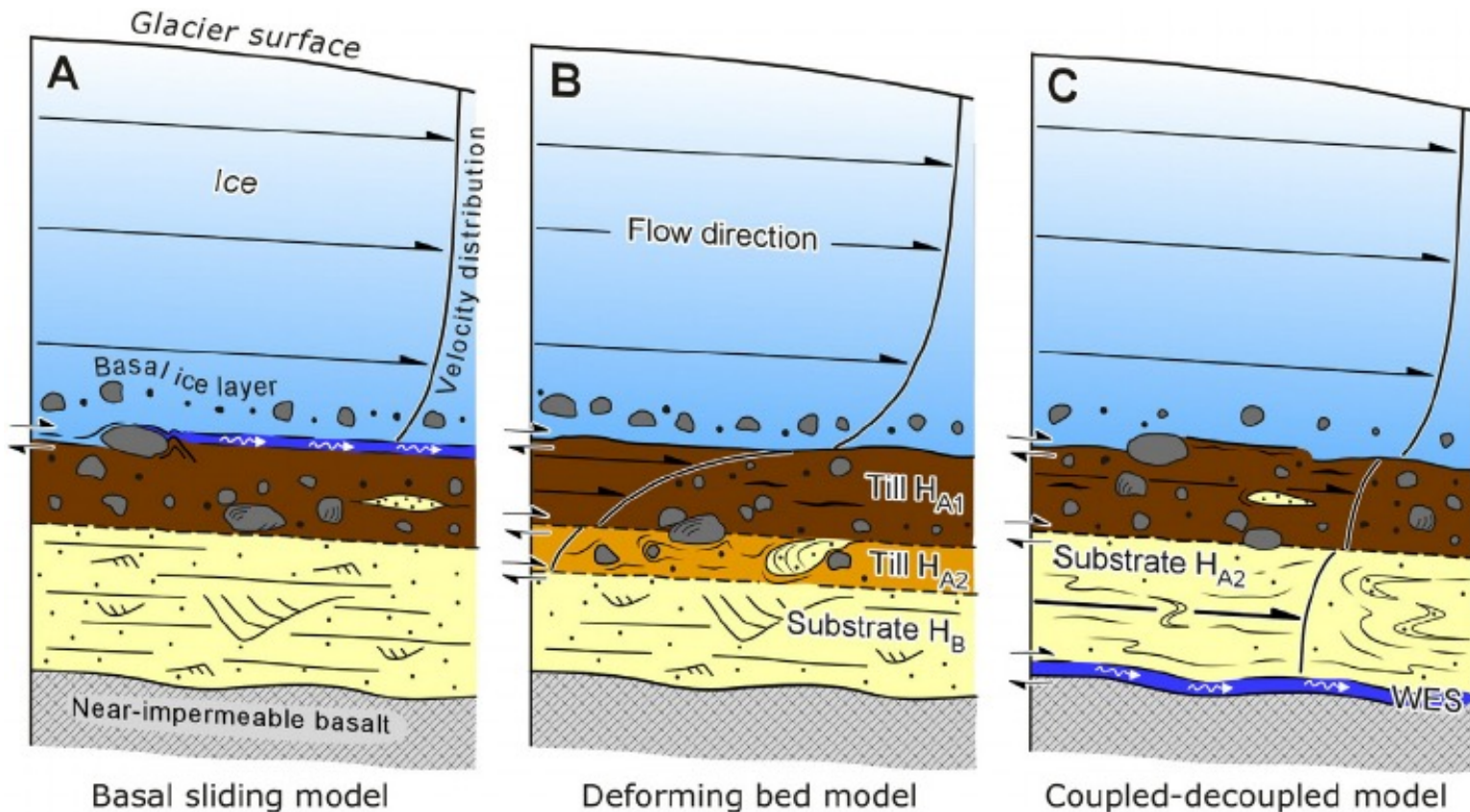


This is a crude model, but it captures some essence of the processes.

Extension of the theory to include cavitation: Iken (1981)

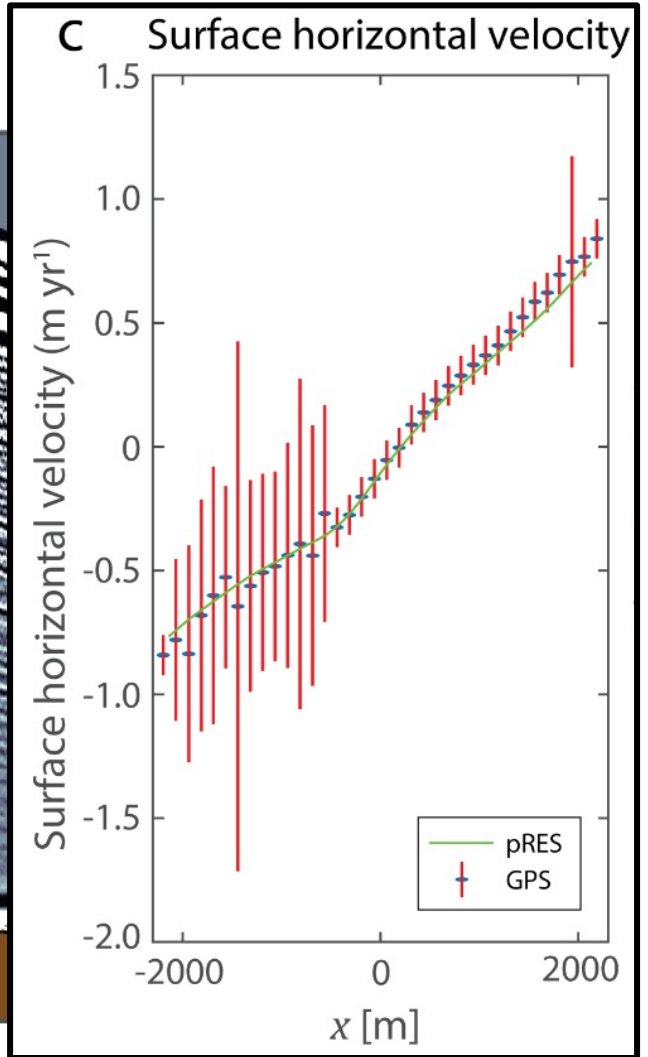
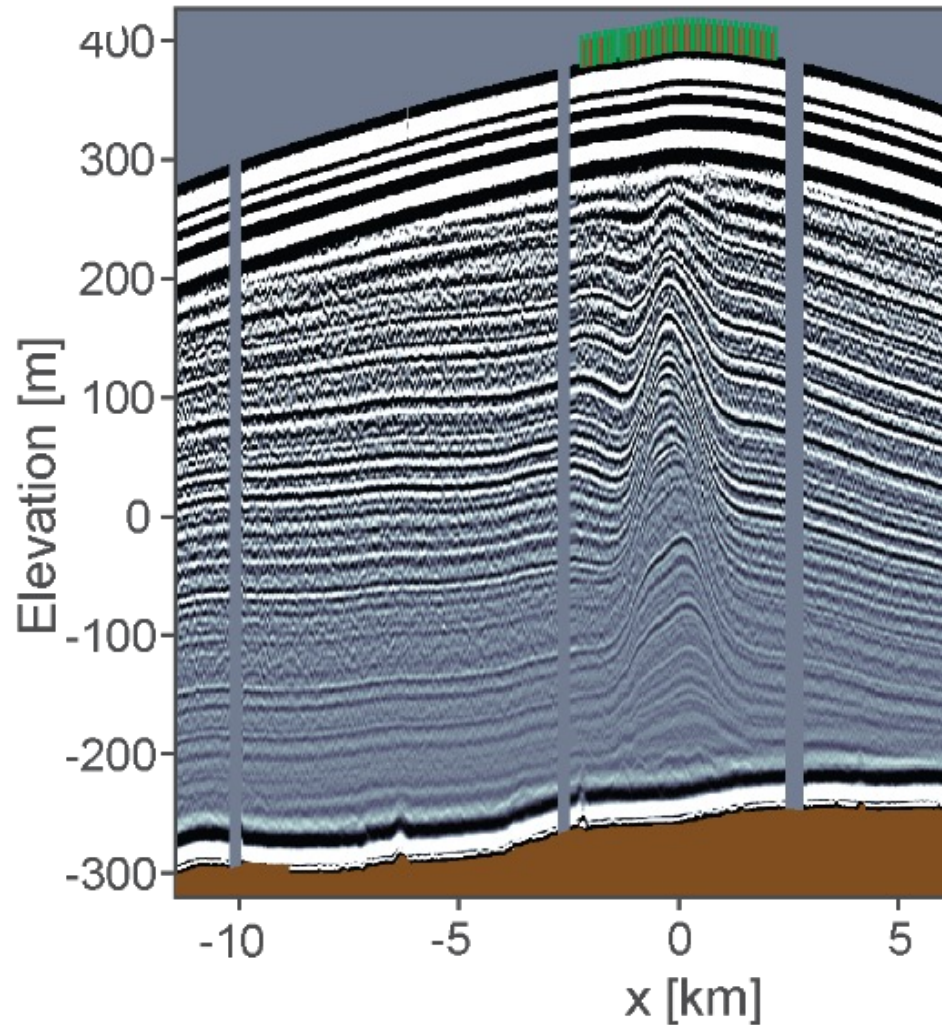
“Soft beds”

deformable material under the ice.



Modified from Kjær et al. (2006)

Ice divides

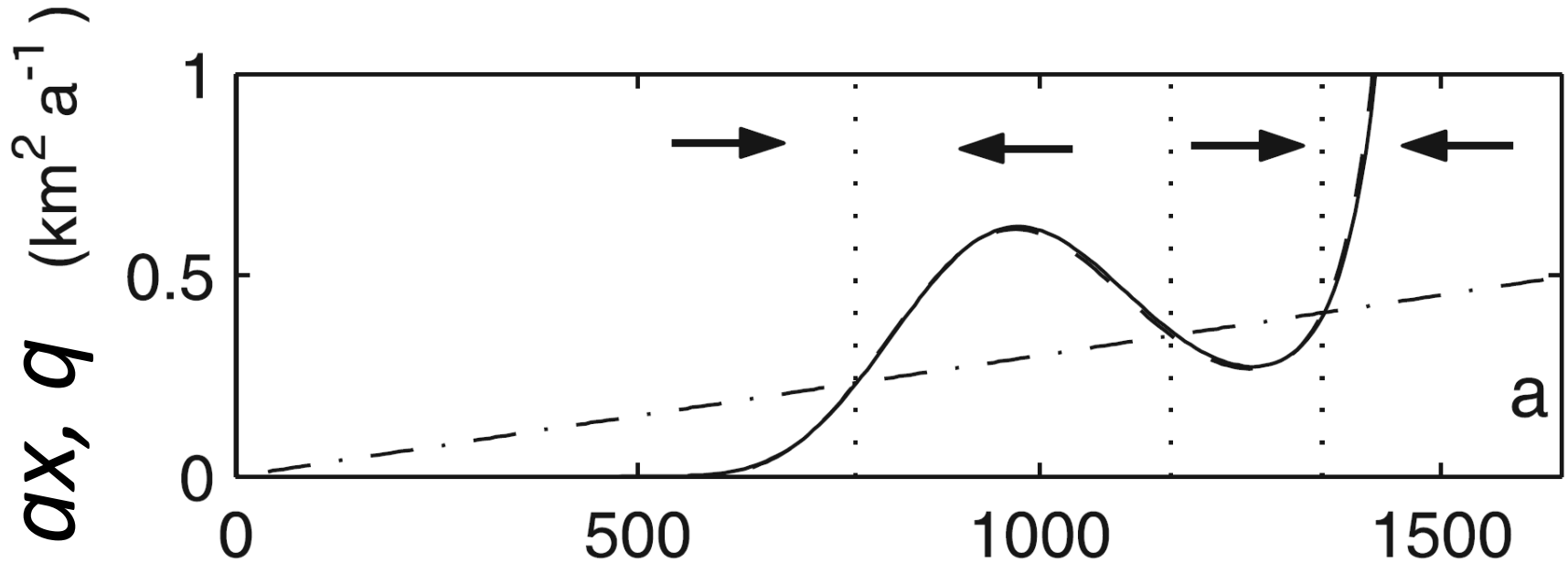
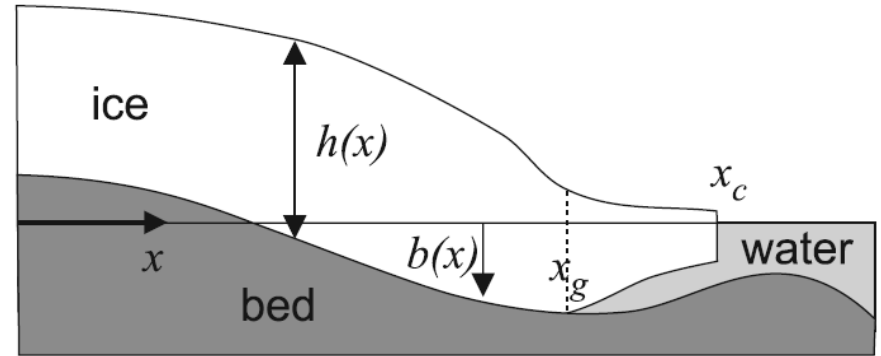


Kingslake et al. (2015)

Schoof's (2007) theory

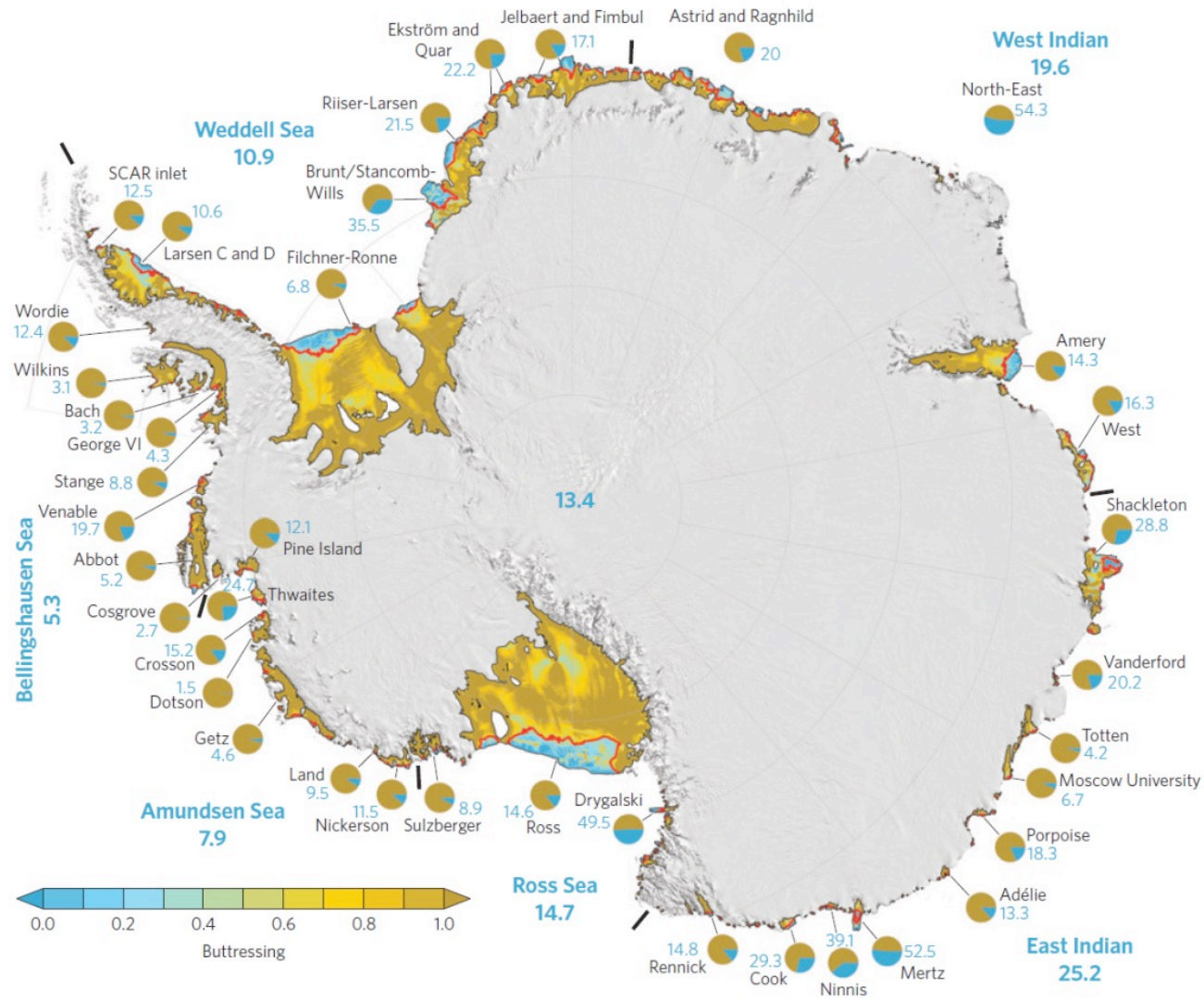
$$q(x_g) \propto h(x_g)^{4.75}$$

$$q = ax$$



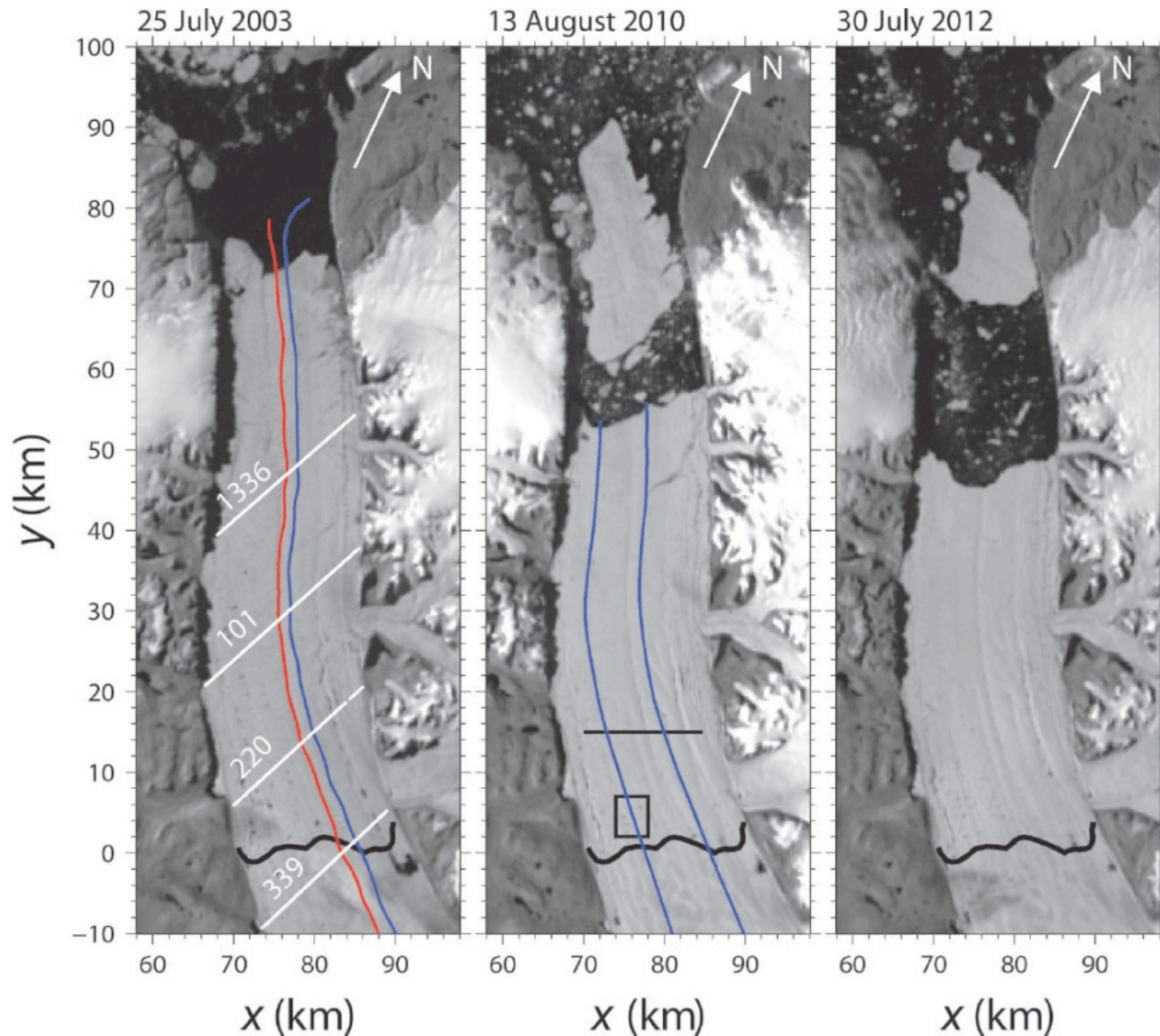
Ice shelves

Antarctic ice shelves



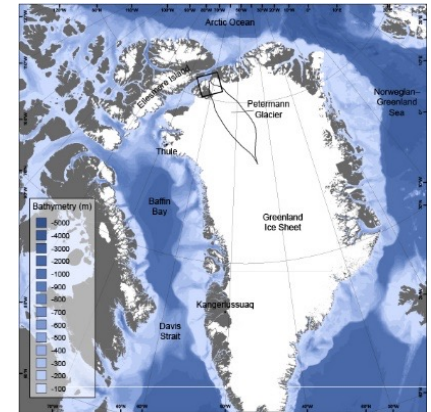
Fürst et al. (2016)

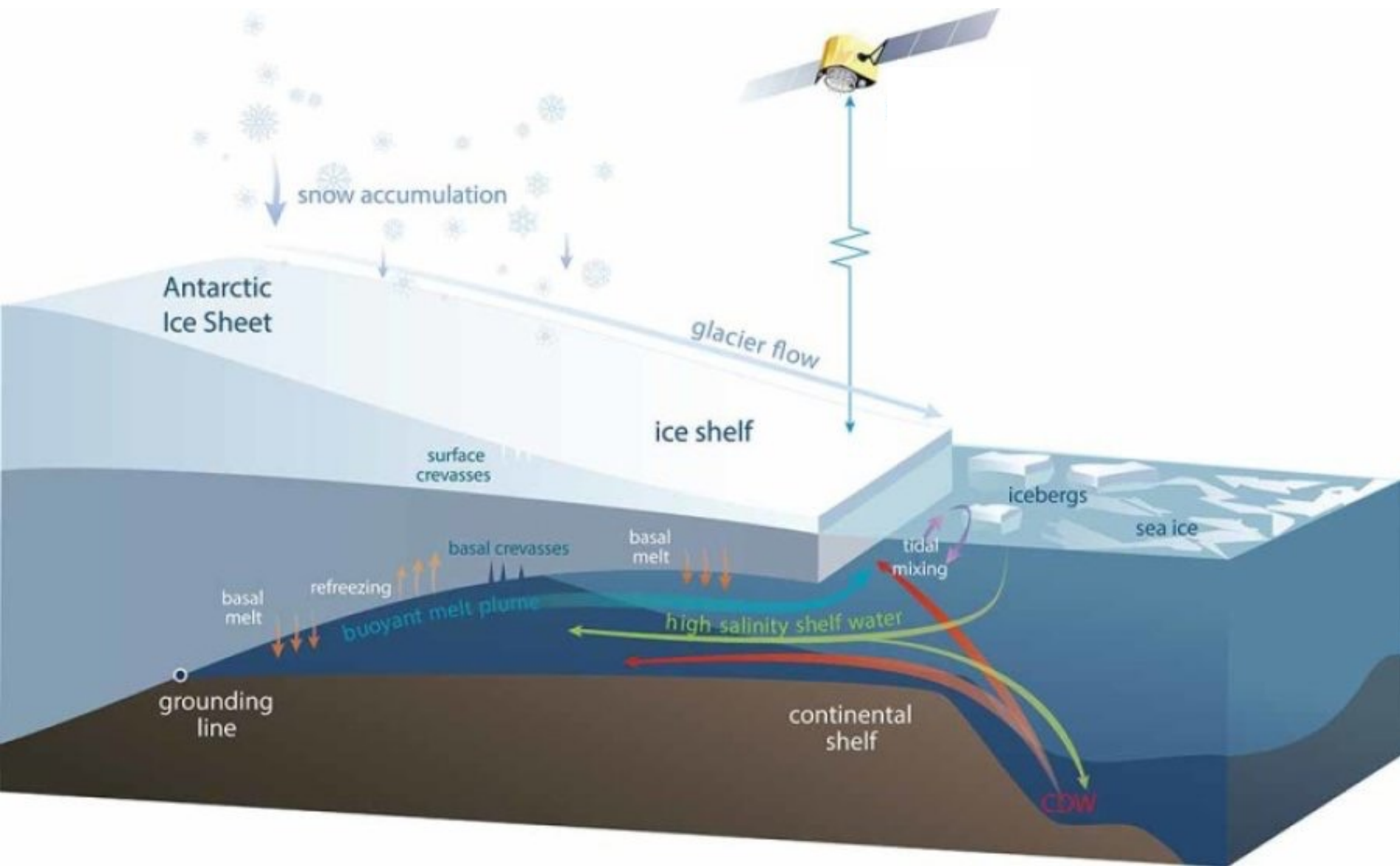
Greenlandic ice shelves



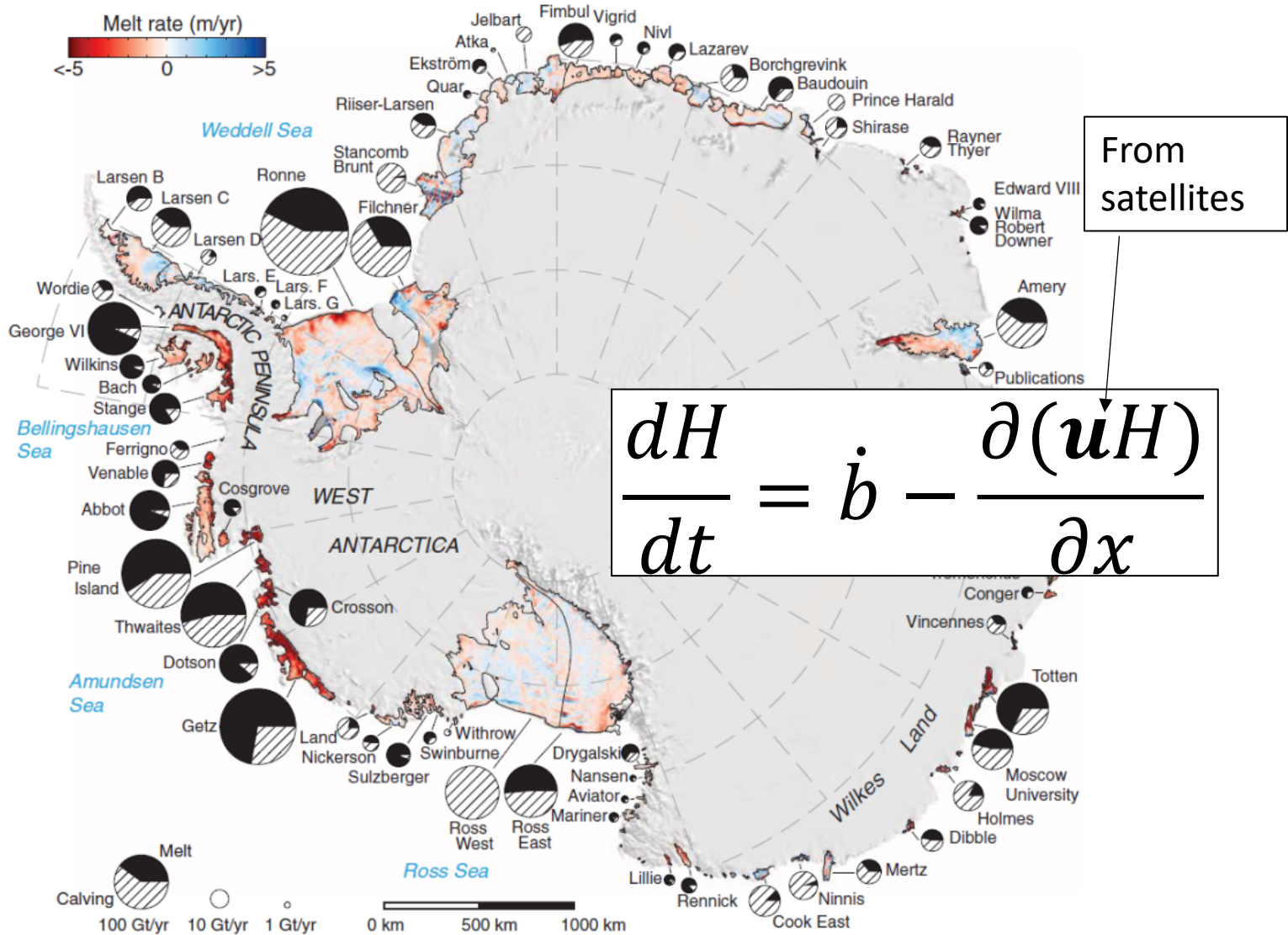
Münchow et al. (2014)

Ice Shelves + review





Ice Shelf mass balance



Rignot et al. (2013)

$$\frac{dH}{dt}$$

and

H

from satellite
altimetry.

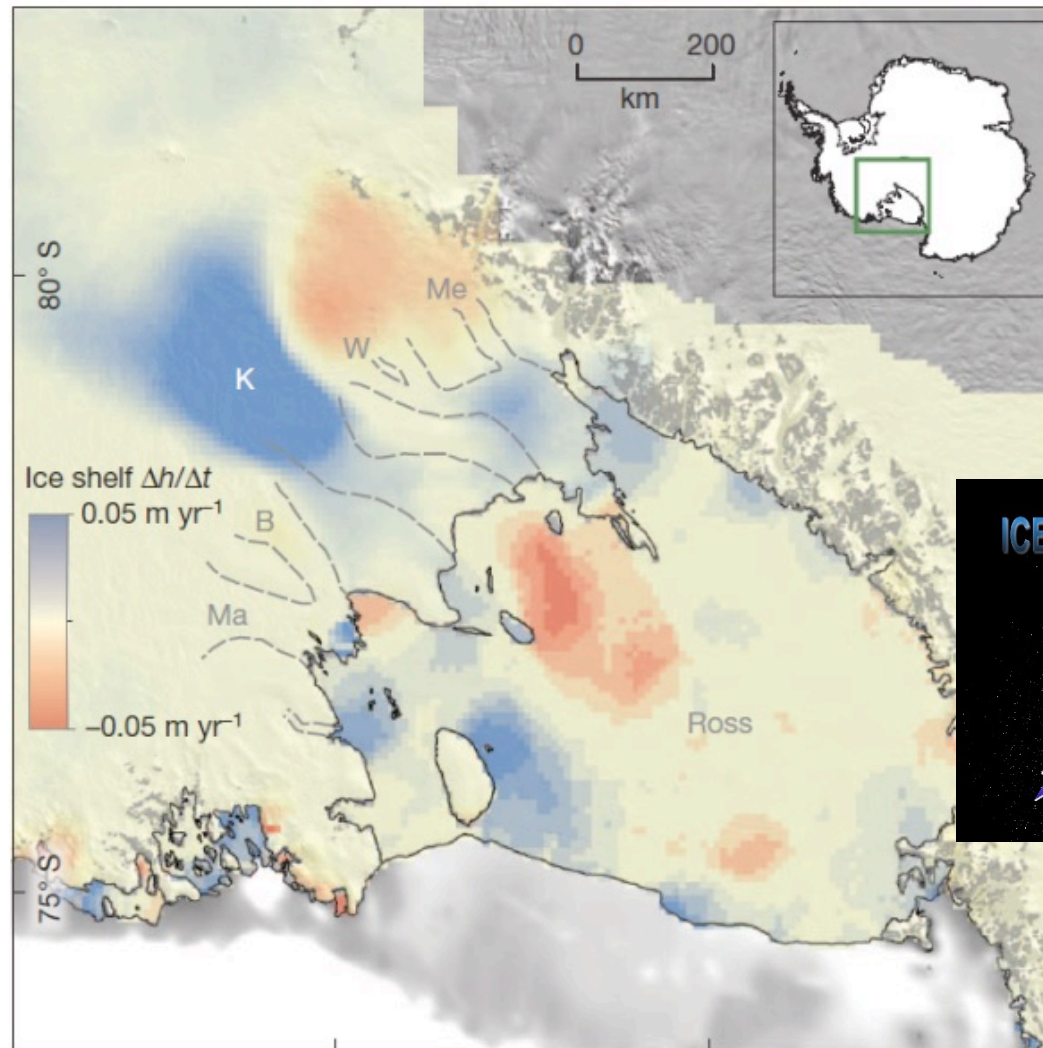
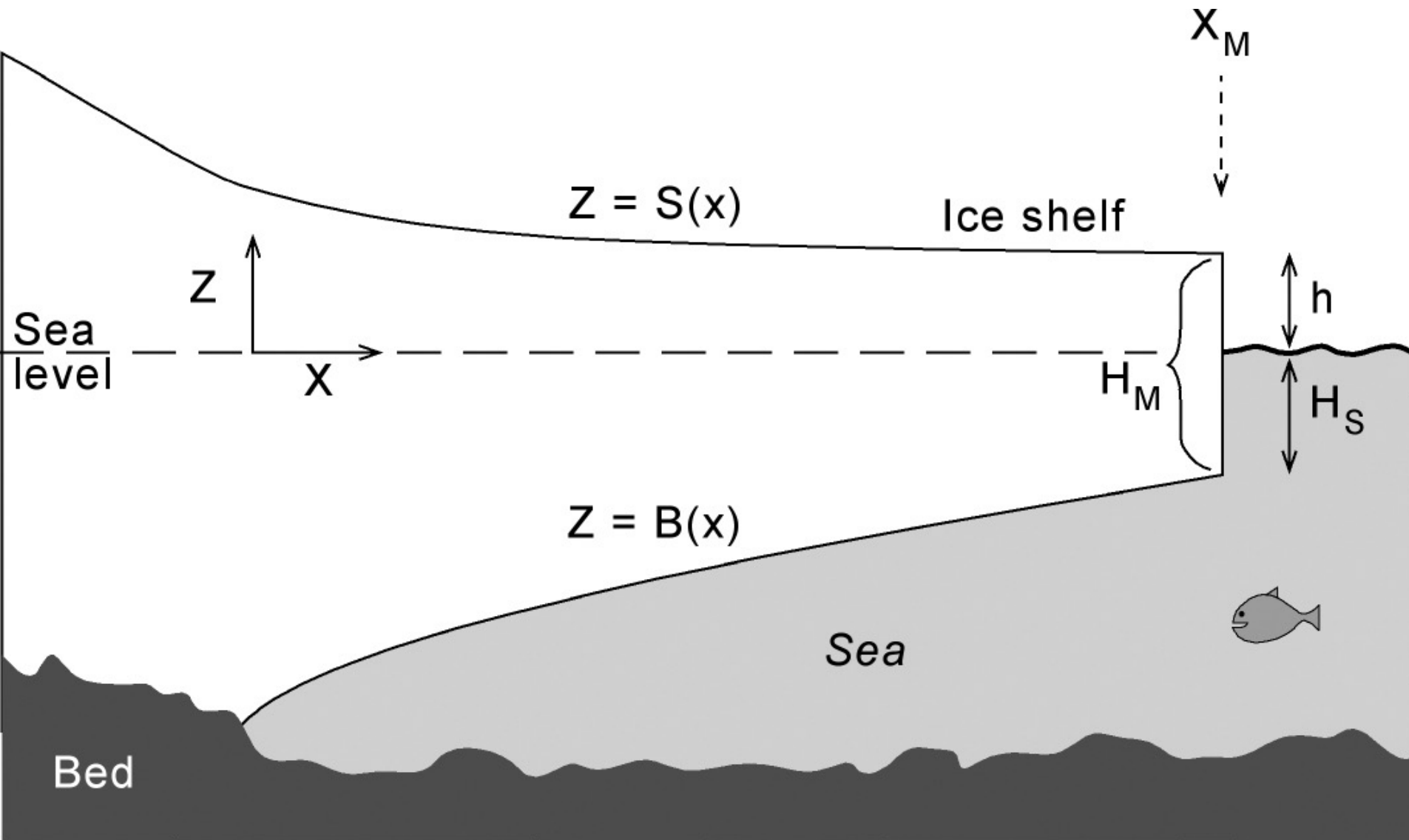


Figure 1 | Surface $\Delta h/\Delta t$ of Ross Ice Shelf, 2003–2008.



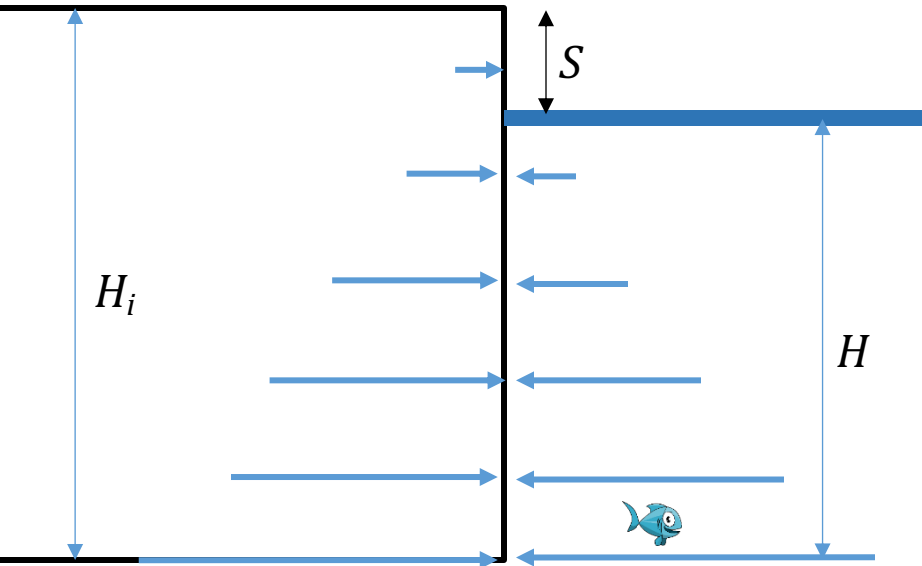
Pritchard et al. (2012)

Ice shelf dynamics



Ice shelf dynamics

Driving stresses



Hydrostatic equilibrium

$$\rho_i g H_i = \rho_w g H$$

$$H = \frac{\rho_i}{\rho_w} H_i$$

There are two sources of driving stress:

1. Surface slope (same as for grounded ice)

$$\rho_i g H_i \frac{dS}{dx}$$

2. Stress imbalance at the front

$$= \int_0^{H_i} \rho_i g z \, dz - \int_0^H \rho_w g z \, dz$$

$$= \frac{1}{2} \rho_i g H_i^2 - \frac{1}{2} \rho_w g H^2$$

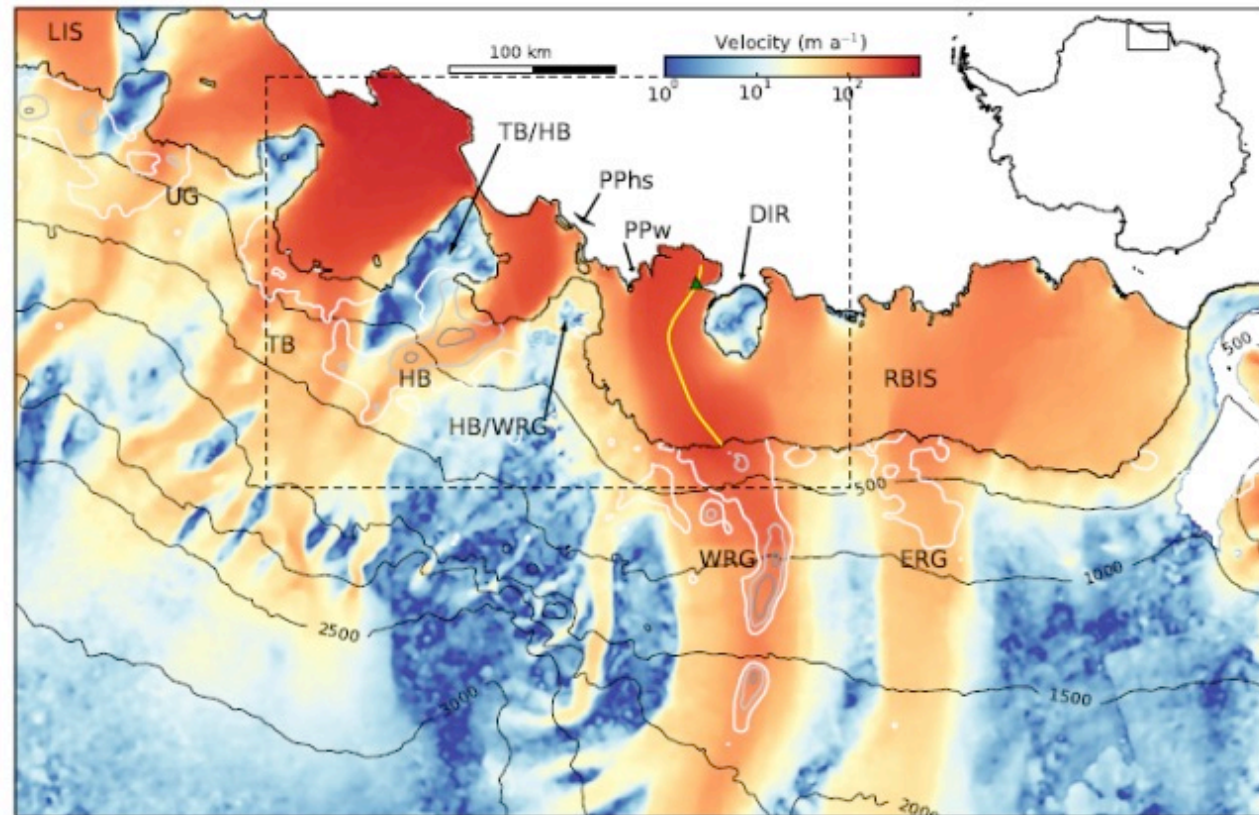
$$= \frac{1}{2} \rho_i g H_i^2 \left(1 - \frac{\rho_i}{\rho_w} \right)$$

Ice shelf dynamics

Resistive stresses

There are three sources of resistive stress:

1. Extensional stresses in the ice
2. Lateral shear stresses
3. Basal shear stress where the ice temporarily grounds.



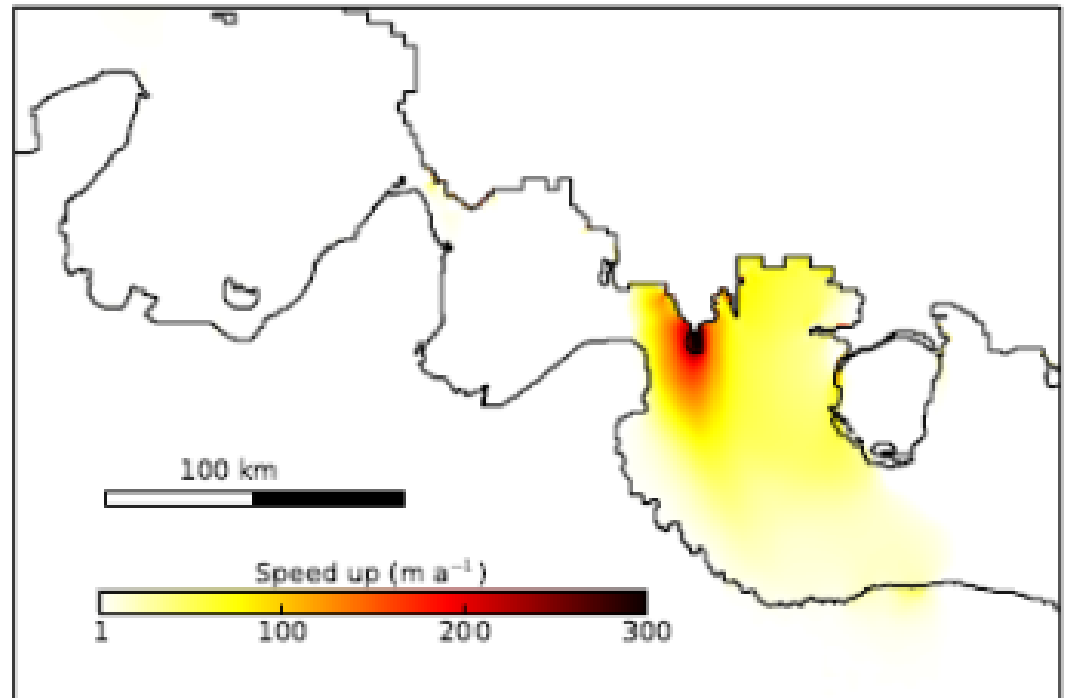
Favier et al. (2016)

Ice shelf dynamics

Resistive stresses

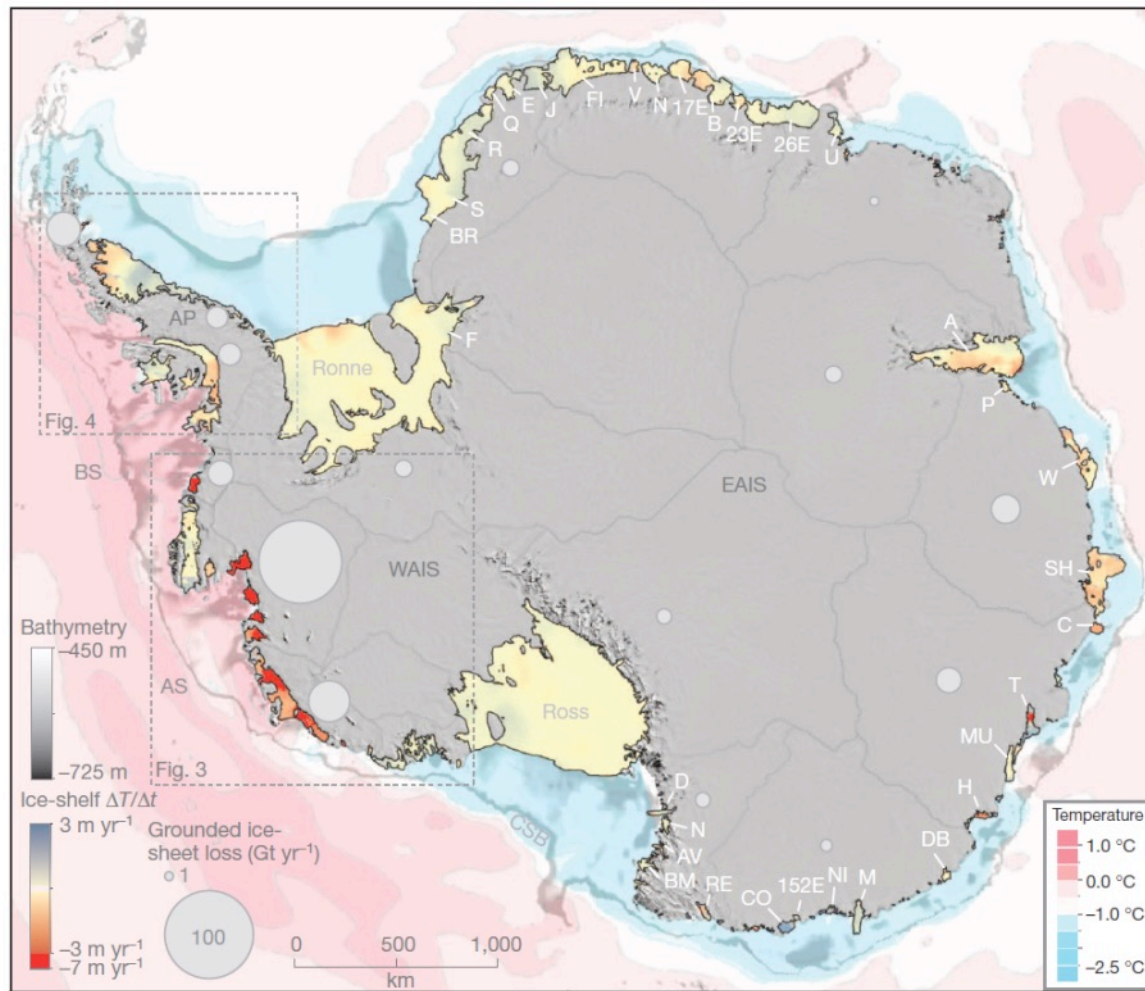
There are three sources of resistive stress:

1. Extensional stresses in the ice
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Favier et al. (2016)

Ice shelf mass loss: thinning

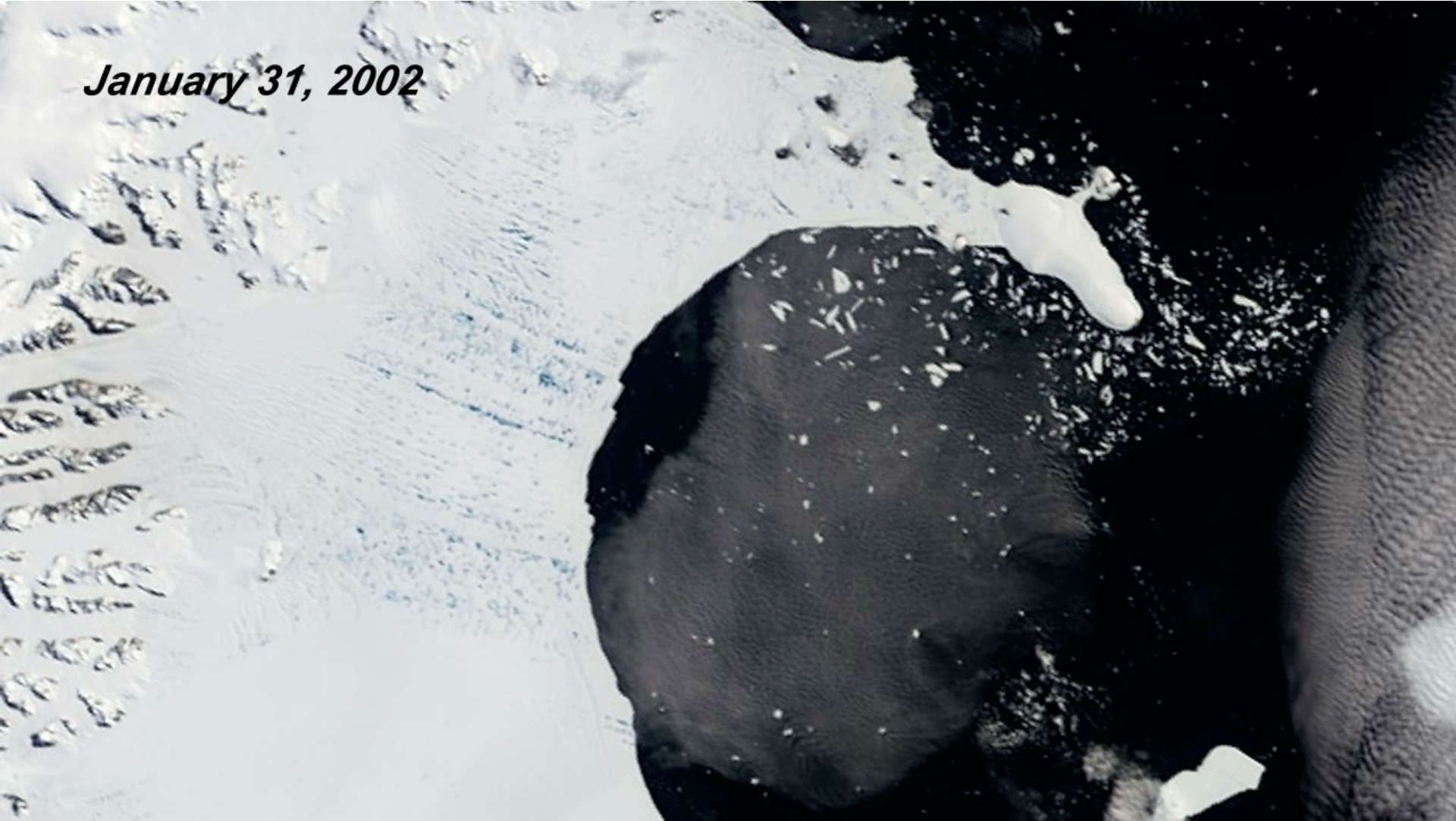


Pritchard et al. (2012)

Ice shelf mass loss: collapse

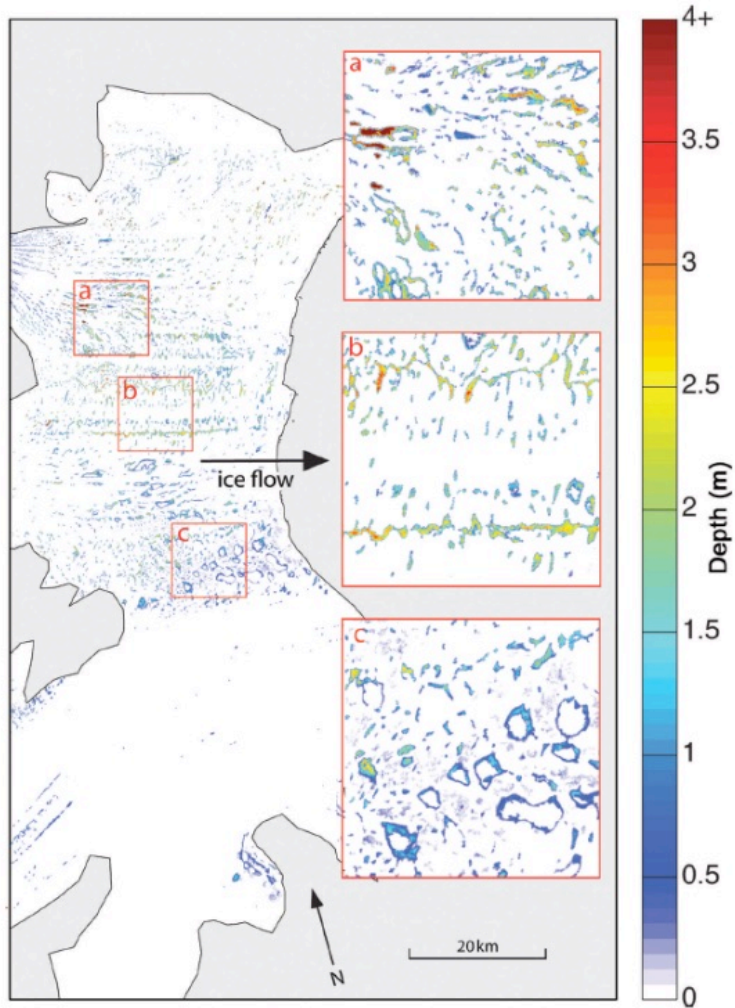
Larsen B Ice Shelf

January 31, 2002

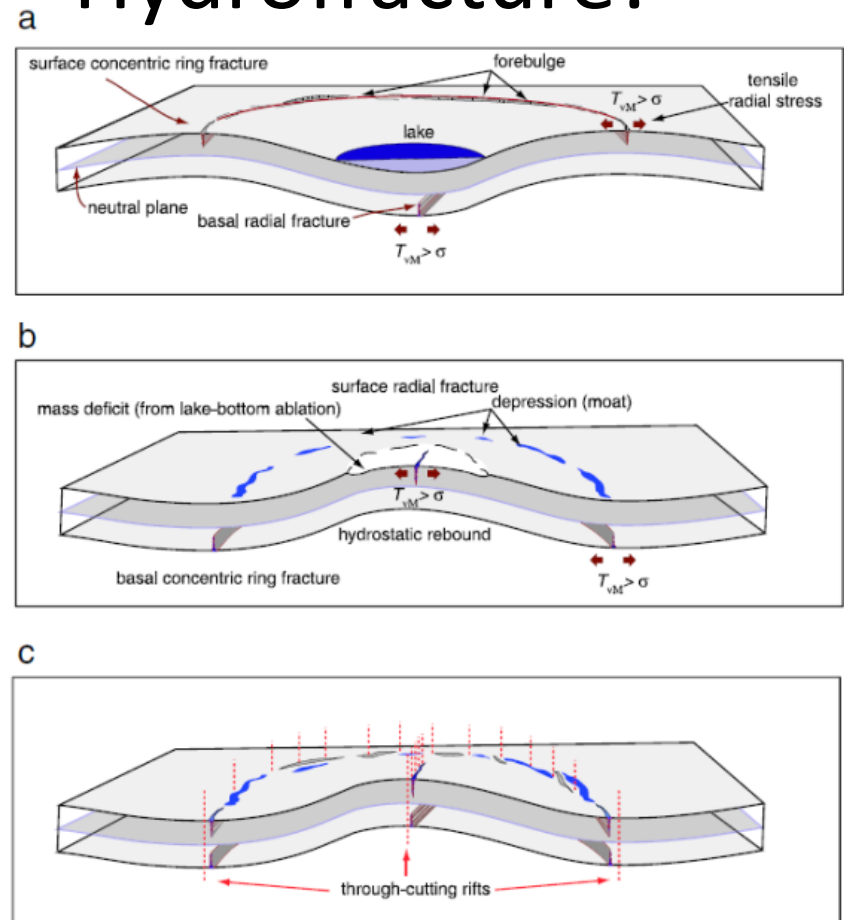


Causes of ice shelf collapse

Hydrofracture!

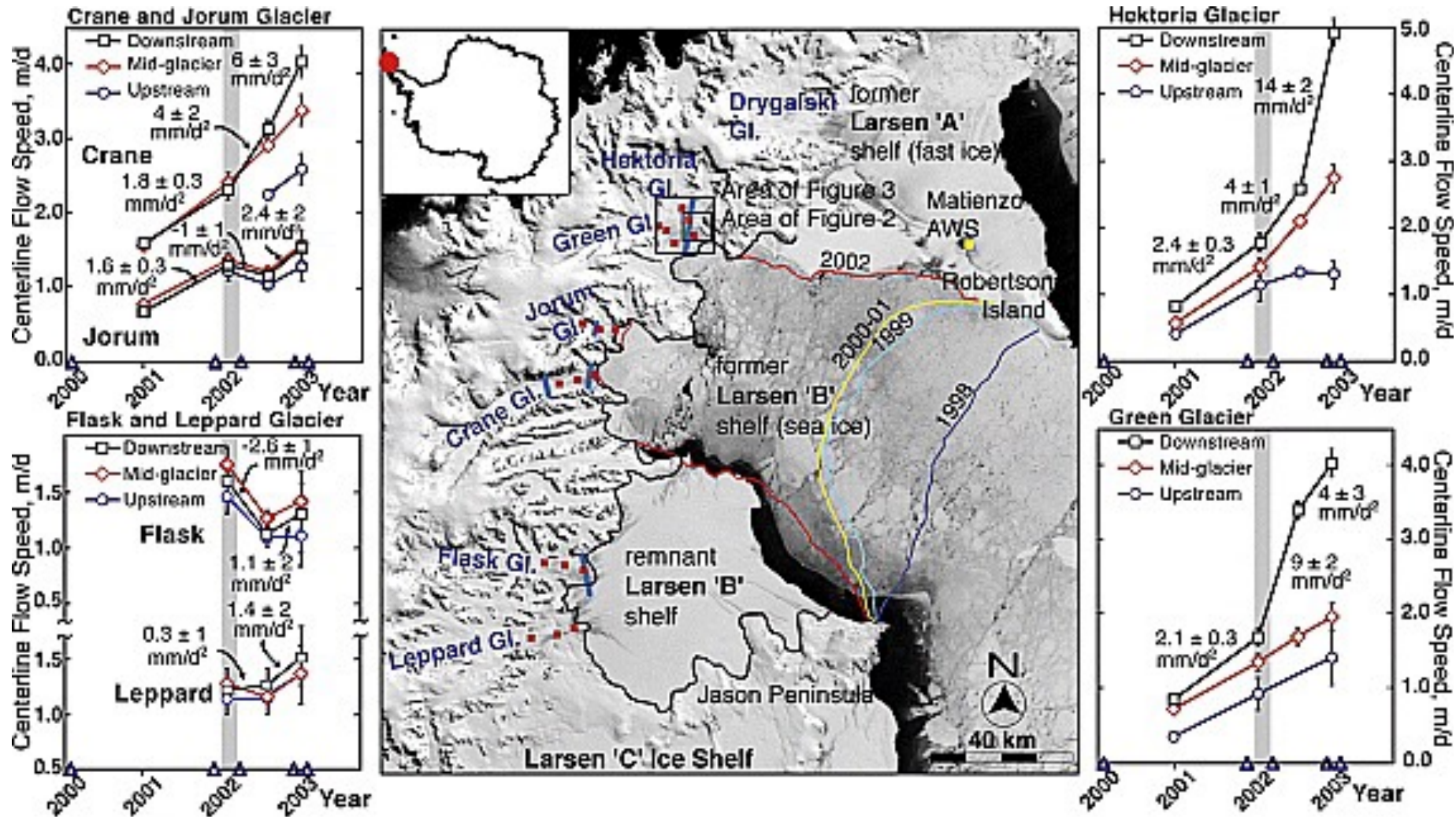


Banwell et al. (2014)



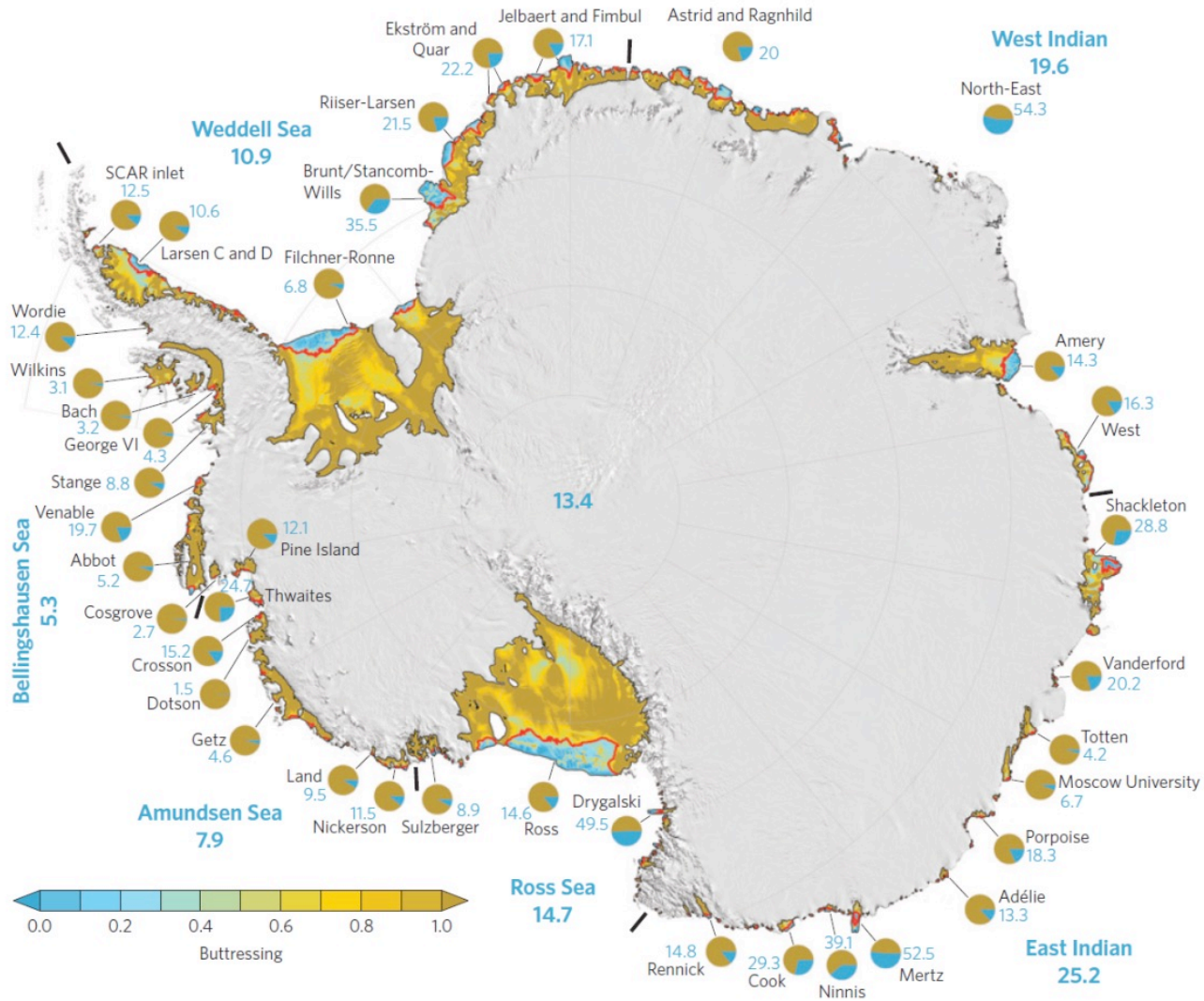
Banwell et al. (2013)

Implications of ice shelf loss



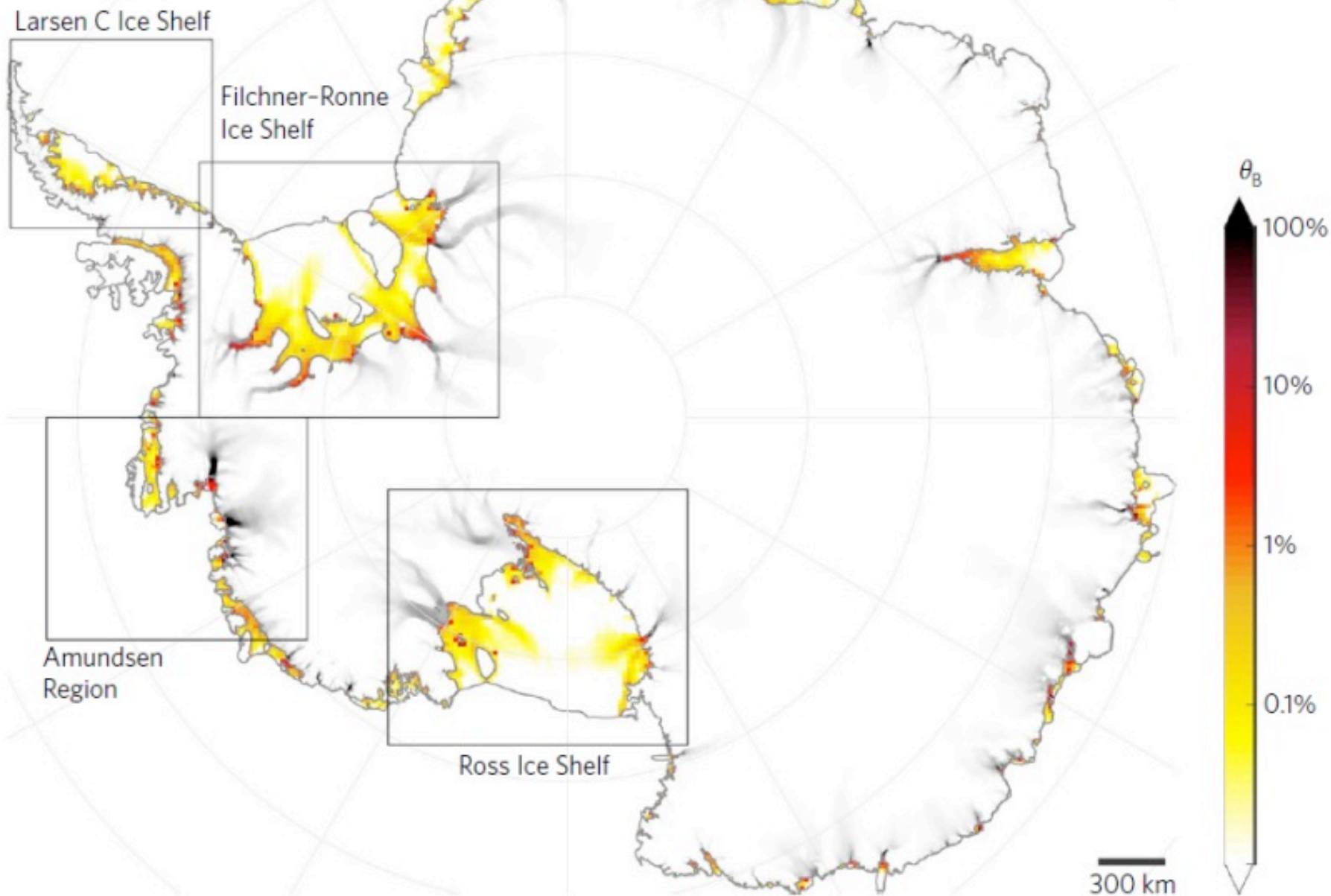
Scambos et al. (2004)

Implications of ice shelf loss



Fürst et al. (2016)

a



Reese et al. (2017)

Ice Shelves + review

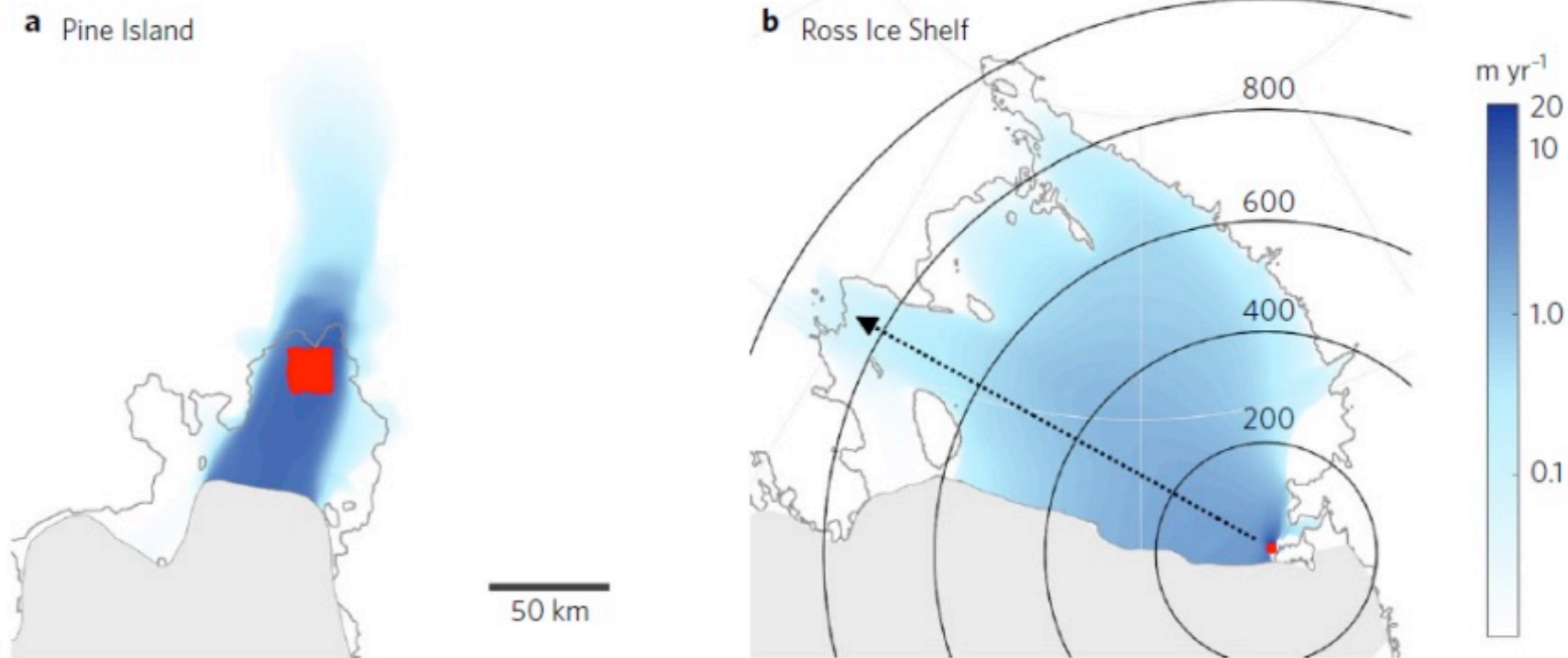


Fig. 2 | Examples of changes in speed resulting from 1 m thinning.

Summary

- Ice shelves are common in Antarctica, rarer in Greenland.
- Satellite-based observations suggests Antarctic ice shelves lose half their mass via basal melting, half from calving.
- Driving forces are due to (1) surface slope and (2) stress imbalance at the front (both would be zero if $\rho_i = \rho_w$).
- Resistive forces are from shear against embayment sides and extensional stresses within ice shelves.
- When ice shelves lose mass or collapse completely, ice upstream accelerates.
- Ice shelves are thinning due to increased ocean melting.
- Collapse appears to be related to surface meltwater through processes such as hydrofracturing.

Next half of semester:

- Heat flow
- Hydrology
- Isochrones and age structure
- Start thinking about your projects

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